European Polymer Journal 223 (2025) 113656

Contents lists available at ScienceDirect

European Polymer Journal

journal homepage: www.elsevier.com/locate/europolj

Direct synthesis of α-Carboxyl-ω-Hydroxyl polymers by catalyst and initiator in the same molecule

Jie Zhang [1], Siheng Zhang [1], Jianda Niu, Zhixian Dong, Jinbao Xu *, Tingting Cui *, Caihong Lei

Guangdong Provincial Key Laboratory of Functional Soft Condensed Matter, School of Materials and Energy, Guangdong University of Technology, Guangzhou 510006, China

ARTICLE INFO

Keywords:
CINICAT
4-hydroxymethyl benzoic acid
α-carboxyl-ω-hydroxyl telechelic polymers

ABSTRACT

With the increasing demands on healthy and environmental issues in modern society, green chemistry and atomic economics have become the consensus of researchers and manufacturers, forcing us to explore much simpler approaches to producing widely utilized biodegradable and biocompatible polyesters. In this context, 4-(hydroxymethyl) benzoic acid (4-HBA), an organic weak acid involving carboxyl and hydroxyl groups, was selected to proceed the ROP of various cyclic monomers, such as ε-caprolactone (CL), δ-valerolactone (VL), l-lactide (LA) and trimethylene carbonate (TMC), as catalyst and initiator in the same molecule (CINICAT) to give α-carboxyl-ω-hydroxyl polymers with controlled molar mass and narrow dispersity (around 1.15). MALDI-TOF, 2D-NMR and other characterizations were applied to confirm the structure of the produced polymers, while density of functional theory simulation (DFT) calculation was carried out to study the structural influence of CINICAT on the polymerizations as well as the polymerization mechanism.

1. Introduction

Considering the depletion of various natural resources (e.g., crude oil and coal) and the destruction of environment (e.g., the long-term stability of commodity plastics) at an ever-increasing rate over the last several decades, chemists in polymer science have been heavily focused on the utilization of more sustainable raw materials and the production of degradable and recyclable functional polymers [1–4]. Several polyesters, particularly the mostly studied poly(ε-caprolactone) (PCL) and poly(lactide) (PLA), which can partially solve the issues mentioned above, have been widely used in the areas of package (e.g., plastic bags and straws) and biomedicine (e.g., degradable sutures and drug delivery carriers) due to their excellent degradability and biocompatibility [4–7]. However, the application of PCL or PLA is usually limited due to its inherent hydrophobic of the backbone, slow degradation rate and lack of functionalities. Consequently, various methods (e.g., chemical modification and blending) have been applied to regulate the properties of PCL or PLA-based materials [8–10].

One of the chemical modification strategies is the incorporation of chain-end functionality, which allows the introduction of expected functional extremity into the macromolecules while leaving the properties of the polymer matrix unchanged. Remarkably, the degradation, thermal stability, self-assembly, and ability to enhance stability of nanoparticles strongly depend on the chain-end functionalities of the polymers [11–13]. In addition, chain-end modification is of particular importance for the biomedical application of PCL or PLA-based materials, such as in drug delivery system, since the drug is usually covalently bound to the polymer matrix through the reaction between the carboxyl or hydroxyl groups of drugs and functional groups of polymers [14–16]. It is well-known that PCL and PLA are predominantly prepared by the ring-opening polymerization (ROP) of cyclic lactones with protic initiators, including alcohols, amines, carboxylic acids, amides, phenols, and thiols, in the presence of metal-based (such as metal alkoxides and alkali metal carboxylates) or organic catalysts (e.g., carbenes, phosphazenes, guanidines, amidines, carboxylic acids), generally only affording monofunctional or telechelic hydroxyl-terminated PCL and PLA [17–22]. Additionally, given the plentiful and varied modifiable sites within drug molecules or organism, it is necessary to diversify the end-group functional groups of PCL or PLA to match this variety. In light of this, the monofunctional or telechelic PCL or PLA that was commonly used has proven insufficient for many emerging functional demands. Consequently, the preparation of telechelic polyesters is of high interest. In

* Corresponding authors.
E-mail addresses: xujinbao@gdut.edu.cn (J. Xu), cuitt@gdut.edu.cn (T. Cui).
[1] These authors contributed equally to this work.

https://doi.org/10.1016/j.eurpolymj.2024.113656
Received 8 September 2024; Received in revised form 12 December 2024; Accepted 14 December 2024
Available online 16 December 2024
0014-3057/© 2024 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

general, there remain two ways to produce end-functional polyesters [12]. The first option is the post-polymerization modification of the terminated hydroxyl groups in polymers. However, the end group functionality is difficult to fully guarantee using this strategy due to the unapproachable quantitative conversion of either in chain end coupling (such as 'click chemistry') or in protection-deprotection methods (e.g., the protection of amine by *t*-butyloxy carbonyl (Boc) and subsequent deprotection with trifluoroacetic acid), which are also unsatisfactory in view of the tedious processes and the possible degradation of the polymer backbone [23–26]. Attractively, the second option can overcome these disadvantages by using hetero-functional initiators in the initial ROP process, which can ensure the final functionality with high accuracy and efficiency. It should be noted that the catalysts employed in ROP are capable of activating protons in the above-mentioned initiators, which generally results in the ROP of monomers being initiated from both sites of hetero-functional groups, without the expected telechelic chain-ends remain intact [27–29]. In this context, the design and preparation of α, ω-telechelic polyesters in a simple manner still seems to be a major challenge.

Zhao et al. reported a noncovalent protection strategy for the one-step synthesis of α-amino, ω-hydroxyl hetero-functional poly(ethylene oxide) (PEO), capitalizing on the noncovalent interactions between triethylborane (Et_3B) and an amino alcohol initiator under mild reaction conditions [30]. Inspired by this, α-amino, ω-hydroxyl telechelic PLA was directly synthesized in our lab in a similar way and PLA-*b*-poly (amino acid)s block copolymers were subsequently prepared in high fidelity using the as-prepared PLA-NH_2 macroinitiator (manuscript pending). Although the noncovalent protection method could directly synthesize telechelic polymers, the usage of toxic, flammable and instable Et_3B is also outside green chemistry. Considering that catalyst residues (either metal or metal free catalysts) in the final products would limit the application of the polyesters (e.g., residual catalyst could can accelerate degradation and reduce the thermal stability of polyesters [31]) and the above discussions, we envision that incorporating the catalyst and initiator into one molecule not only avoid the tedious catalyst removing procedure but also enables the preparation of α, ω-telechelic polymer since the ROP of CL works well with an organic acid (under certain circumstances serve as a functional chain end) catalyst [32–34]. For example, Liu et al. reported the ROP of CL by some natural amino acids and the resulting PCL was initiated from the amine, affording PCL with α-carboxyl-ω-hydroxyl, but the description of the polymerization kinetics and further analysis of the hetero-functionality were insufficient [35]. Recently, Penczek el al. found that well-defined α-phosphoric acid, ω-hydroxyl hetero-functionalized PCL, PLA and poly(trimethylene carbonate) (PTMC) can be directly synthesized by the cationic ROP in high efficiency when the catalyst and initiator are in the same molecule (CINICAT) [36–39]. This stands in contrast to the traditional cationic ROP processes, where the use of organocatalysts such as protonic acids requires the incorporation of a separate initiator, resulting in the presence of a low molecular weight catalyst in the polymer product (the catalyst residues are harmful for later biomedical application).

Having these in mind, a commercially available weak acid 4-(hydroxymethyl) benzoic acid (4-HBA) (used in the synthesis of disease inhibitors, hydrophobic peptides,[40] etc.), within the hydroxyl group (initiator) and carboxyl group (catalyst) in the same molecule, which could also be viewed as a kind of CINICAT, was selected to synthesize α-carboxyl-ω-hydroxyl telechelic PCL in a straightforward way (Scheme 1) taking into account the (expected) research or application of carboxyl-terminated polyesters in the area of crystallization, drug carrier design, self-assemble, and degradation. In particular, we performed (i) the characterization of the polymers synthesized from CINICAT, (ii) the polymerization kinetics and chain-extension experiment and (iii) the structural influence of CINICAT on polymerizations as well as the polymerization mechanism based on the density of functional theoretical simulation (DFT) calculations.

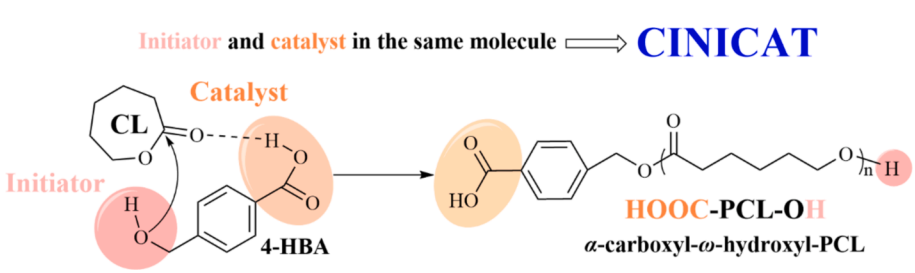
2. Experiments

2.1. Materials

Aladdin-sourced CL (99 %) and δ-valerolactone (VL, 99 %) were combined with calcium hydride (CaH_2) and stirred continuously overnight. Subsequently, the mixture underwent vacuum distillation twice, with the product being stored in a glove box. Trimethylene carbonate (TMC, Aladdin, 98 %) and L-lactide (LA, Sigma-Aldrich, 98 %) were each recrystallized twice using anhydrous tetrahydrofuran (THF) and ethyl acetate, respectively, which were then dried under vacuum at 40 °C overnight and then stored at − 10 °C within a glove box. 4-HBA, 2-HBA, 3-HBA, 2-fluoro-4-hydroxymethyl-benzoic acid (2-F-4-HBA) and 4-hydroxymethyl-3-methoxy-benzioc acid (3-M−4−HBA) (Sigma-Aldrich, ≥99 %) were dried at 80°C overnight under vacuum (stored in a glove box). Reagents sourced from Aladdin and Sigma-Aldrich were utilized in their original, unaltered form.

2.2. Characterizations

A Bruker AV400 NMR spectrometer was used to analyze the samples for nuclear magnetic resonance (NMR). Deuterated chloroform ($CDCl_3$) or acetonitrile (CD_3CN) containing internal standard of tetramethylsilane (TMS) were used as solvent for ^{1}H NMR, ^{13}C NMR or 2D-NMR spectra (^{1}H–^{13}C HMBC and ^{1}H DOSY NMR) characterizations. The apparent number-average molecular weight, denoted as $M_{n,SEC}$ and dispersity, represented by $Đ$ were determined using a size exclusion chromatography (SEC, Waters). This setup included a 1515 pump and a 2414 differential refractive index detector with THF (Aladdin, HPLC) serving as eluent at a flow rate of 1.0 mL/min at a temperature of 35 °C. Prior to analysis, all samples were dissolved in HPLC-grade THF to a concentration of 5.0 mg/mL. Subsequently, the solutions were filtered through a 0.40 μm hydrophobic membrane. The experimental molecular weights were then deduced from a calibration curve, which was established using a set of monodisperse polystyrene (PS) standards. Matrix-assisted laser desorption ionization time of flight mass spectrometry (MALDI-TOF MS) was employed to assess the precise molecular weight and terminal functionality of the synthesized polymers, using a Bruker ultrafleXtreme MALDI-TOF instrument. The mass spectra were captured in reflector mode, with a total of one hundred shots accumulated at an acceleration voltage of 25 kV. For sample preparation, a solution



Scheme 1. Synthesis of PCL with α-carboxyl-ω-hydroxyl chain ends by CINICAT.

containing the purified polymer (at a concentration of10.0 mg/mL with a volume of 10.0 μL), matrix (2,5-dihydroxybenzoic acid at 15.0 mg/mL, 50.0 μL) and cationic agent (sodium iodide at 4.0 mg/mL, 10.0 μL) was deposited on the sample platform and subsequently dried for the MALDI-TOF MS analysis.

2.3. Synthesis of polyester in the presence of CINICAT

One typical procedure for the ROP of CL in the presence of CINICAT is depicted in detail as follows: CL (0.57 g, 5.0 mmol, 50 equiv.) and 4-HBA (15.2 mg, 0.1 mmol, 1.0 equiv.) were added to a dried Schlenk reactor containing a stirrer (proceeded in a glove box). Subsequently, the reactor was positioned within a glove box and set to a temperature of 150 °C, whereupon aliquots of the polymerization mixture were extracted at the predetermined time intervals. Upon collection, each aliquot was promptly transferred to a refrigerator (−10 °C in a glove box) to stop the polymerization, and then the samples were dissolved for further characterizations (e.g., ^{1}H NMR and SEC). Ultimately, the polymerization process was terminated after a duration of 24 h and dissolved in 3 mL dichloromethane (DCM), followed by precipitation into an excess of cold methanol. The resulting white solid was subsequently dried under vacuum conditions at a temperature of 40 °C overnight for further characterizations. It is a similar process for the polymerization of CL, VL, LA and TMC by CINICAT under different conditions. PCL (yield: 96.7 %, $M_{n,SEC}$ = 6 400 g/mol, $Đ$ = 1.16, Table 1, Table. S3, Fig. 1A), ^{1}H NMR (400 MHz, $CDCl_3$, ppm): 1.33 (2H, –CO$(CH_2)_2CH_2(CH_2)_2$O–), 1.62 (4H, –COCH$_2$CH_2CH$_2$CH_2CH$_2$O-), 2.26 (2H, –COCH_2-$(CH_2)_4$O-), 3.62 (2H, –CO$(CH_2)_4CH_2$OH), 4.06 (2H, –CO$(CH_2)_4CH_2$O-), 5.04 (2H, HOOCC$_6$H$_5$CH_2O-), 7.43 and 8.08 (4H, aromatic); PVL (yield: 98.5 %, $M_{n,SEC}$ = 6 800 g/mol, $Đ$ = 1.11, Table 1, Table. S7, Fig. S1A), 1.66 (4H, –COCH$_2$(CH_2)$_2$CH$_2$O–), 2.32 (2H, –COC$H_2$$(CH_2)_3$O–), 3.64 (2H, –CO$(CH_2)_3CH_2$OH), 4.05 (2H, –CO$(CH_2)_3CH_2$O–), 5.04 (2H, HOOCC$_6H_5CH_2$O-), 7.30 (4H, aromatic); PTMC (yield: 95.0 %, $M_{n,SEC}$ = 7 400 g/mol, $Đ$ = 1.50, Table 1, Table. S8, Fig. S2A), 2.05 (2H, –COOCH$_2$CH_2CH$_2$O-), 3.72 (2H, –COO$(CH_2)_2CH_2$OH), 4.23 (4H, –COOCH_2CH$_2$CH_2O-), 5.05 (2H, HOOCC$_6$H$_5$-CH_2O-), 7.30 (4H, aromatic); PLA (yield: 94.3 %, $M_{n,SEC}$ = 8 200 g/mol, $Đ$ = 1.19, Table 1, Table. S11, Fig. S3A), ^{1}H NMR (400 MHz, $CDCl_3$, ppm), 1.55 (3H, –COCH(CH_3)O-), 4.33 (1H, –COCH(CH$_3$)OH), 5.10 (1H, –COCH(CH$_3$)O-), 7.43 and 8.08 (4H, aromatic).

2.4. Block copolymerization of CL and VL, TMC, LA in the presence of CINICAT

An example has been described in detail here to illustrate the process of block copolymerization of CL and VL. CL (0.57 g, 5.0 mmol, 50 equiv.) and 4-HBA (15.2 mg, 0.1 mmol, 1.0 equiv.) were added to a dried Schlenk reactor involving a stirrer (150 °C in a glove box). An aliquot (for ^{1}H NMR and SEC analysis) was taken out after 20 h and the designed amount of VL (0.5 g, 5.0 mmol, 50 equiv.) was then added to start the copolymerization. Subsequently, the reaction mixture was dissolved in DCM (8 mL) and precipitated into excess of methanol after another 16 h. Ultimately, the isolated white solid was collected and dried under vacuum for further analysis. PCL-*b*-PVL (yield 85 %, $M_{n,SEC}$ = 16 600 g/mol, $Đ$ = 1.35, Table 2, Table. B1). ^{1}H NMR (Fig. S4A, 400 MHz, $CDCl_3$, ppm): 1.36 (2H, –CO$(CH_2)_2CH_2(CH_2)_2$O), 1.66 (4H, –COCH$_2$CH_2CH$_2$CH_2CH$_2$O-, 4H, –COCH$_2$CH_2CH$_2$CH_2CH$_2$O-), 2.32 (2H, –COC$H_2$$(CH_2)_4$O-, 2H, –COC$H_2$$(CH_2)_3$O–), 3.65 (2H, –CO$(CH_2)_4CH_2$OH, 2H, –CO$(CH_2)_3CH_2$OH), 4.06 (2H, –CO$(CH_2)_4CH_2$O-, 2H, –CO$(CH_2)_3CH_2$O–), 5.03 (2H, HOOCC$_6H_5CH_2$O-), 7.43 and 8.08 (4H, aromatic). The copolymerization of CL and TMC, LA in the presence of CINICAT was performed in a similar process. (PCL-*b*-PTMC) (yield 86.7 %, $M_{n,SEC}$ = 16 100 g/mol, $Đ$ = 1.47, Table 2, Table. B2). ^{1}H NMR (Fig. S4C, 400 MHz, $CDCl_3$, ppm): 1.34 (2H, –CO$(CH_2)_2CH_2(CH_2)_2$O–), 1.63 (4H, –COCH$_2$CH_2CH$_2$CH_2CH$_2$O-), 2.04 (2H, –COOCH$_2$CH_2CH$_2$O-), 2.30 (2H, –COC$H_2$$(CH_2)_4$O-), 3.65 (2H, –CO$(CH_2)_4CH_2$OH, 4.06 (2H, –CO$(CH_2)_4CH_2$O-), 4.12 (4H, –COOC$H_2CH_2CH_2$O-), 5.03 (2H, HOOCC$_6H_5CH_2$O-), 7.43 and 8.08 (4H, aromatic). (PCL-*b*-PLA) (yield 89.5 %, $M_{n,SEC}$ = 11 800 g/mol, $Đ$ = 1.31, Table 2, Table. B3). ^{1}H NMR (Fig. S4E, 400 MHz, $CDCl_3$, ppm): 1.38 (2H, –CO$(CH_2)_2CH_2(CH_2)_2$O), 1.55 (6H, –COCH(CH_3)COOCH(CH_3)O-), 1.67 (4H, –COCH$_2$CH_2CH$_2$CH_2CH$_2$O-), 2.36 (2H, –COC$H_2$$(CH_2)_4$O-), 3.74 (1H, –COCH(CH$_3$)COOC$H$(CH$_3$)OH), 4.10 (2H, –CO$(CH_2)_4CH_2$O-), 5.05 (2H, C$_6H_5CH_2$O-), 5.17 (2H, –COC$H$(C$H_3$)COOC$H$(C$H_3$)O-), 7.43 and 8.08 (4H, aromatic).

2.5. DFT calculations

DFT calculations were performed utilizing the DMol^3 module, employing the generalized gradient approximation (GGA) with Perdew – Burke – Ernzerhof (PBE) functional for the exchange–correlation term [41,42]. The computational basis set was set to double numerical plus polarization (DNP), and the core treatment included all electrons with

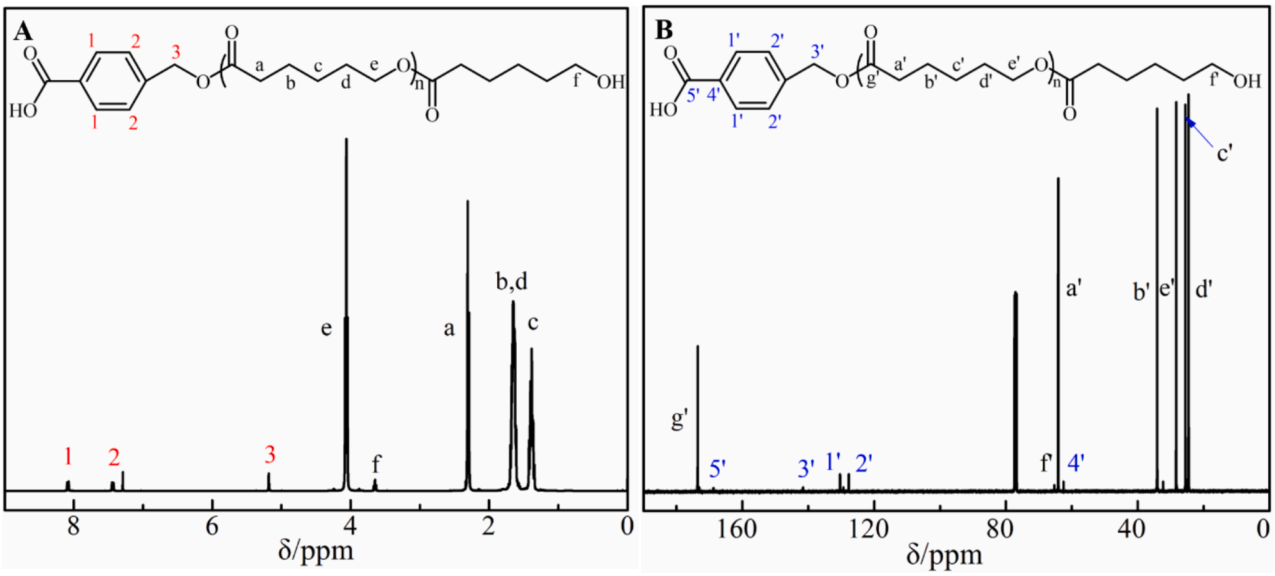


Fig. 1. (A) ^{1}H NMR and (B) ^{13}C NMR spectra of the isolated PCL in the presence of 4-HBA (Table 1, Table. S3).

some relativistic effects. The DFT-D method within the Grimme scheme was selected to account for van der Waals interactions. The geometric structures were fully optimized until the forces, displacements, and energy changes were below the thresholds of 0.002 0.002 Ha/ Å (1 Ha = 27.211 eV), 0.005 Å and 1×10^{-5} Ha, respectively. Linear synchronous transition/quadratic synchronous transition (LST/QST) and nudged elastic band (NEB) methodologies were applied to analyze the dissociation and recombination pathways for the CINICAT catalyst.

3. Results and discussion

3.1. Determination the structure of the isolated PCL in the presence of CINICAT

Given the simplicity, cost-effectiveness, and scalability of bulk polymerization, 4-HBA was used to catalyzed the ROP of CL, VL, TMC and LA in bulk (Scheme 1). The results are summarized in Table 1. The ROP of CL was first investigated at 80 °C considering the melting temperature of PCL ($T_m \approx 60$ °C) and its general selection in other publications,[20,43] but the monomer conversion after 24 h is particularly low (Table 1, Table. S1). We speculate that the acidity of 4-HBA is not strong enough and increasing the reaction temperature might be an option [33,44]. Therefore, the ROP temperature was then increased to 130 °C and the monomer conversion reached 80.7 % within 24 h (Table 1, Table. S2), which is still unsatisfactory. Excitingly, a monomer conversion of 96.7 % (Table 1, Table. S3) was achieved after 24 h when the temperature was increased to 150°C. In addition, benzyl alcohol, a commonly used initiator in the ROP of CL, was mixed with CL (Table 1, Table. S12) at 150 °C for 24 h resulting in only 5 % monomer conversion, demonstrating the dual role of 4-HBA in the ROP process. The ^{1}H NMR (Fig. 1A) of the purified PCL shows the typical peaks of methylene from the backbone of PCL and the signals at 7.43, 8.08 and 5.04 ppm are attributed to the aromatic and methylene protons of 4-HBA, respectively, while the signals in ^{13}C NMR (Fig. 1B) also clearly shows the presence of PCL and 4-HBA compartments in the final product. Furthermore, the molecular weight of the isolated PCL from SEC is consistent with the theoretical and NMR values, and the narrow value of Đ implies that the polymerization is controllable in this catalytic system. It is noticeable that we are just in the process of confirming the successful ROP of CL by CINICAT from the results of NMR and SEC (much more evidence is needed).

To further substantiate the structure of the obtained PCL, MALDI-TOF MS and 2D NMR were subsequently selected due to their high efficiency and accuracy in detecting the end group fidelity and the relationship between the main chain and the chain end [45,46]. It is obvious that the $M_{n,theo.}$ values (Fig. 2) of linear PCL with α-HOOCC$_6$H$_4$CH$_2$O- and ω-H chain end (sodium iodide was used as a cation agent) are in reasonable agreement with the main signals, while the minor peaks could be assigned to the partially ionized carboxyl (the calculated values) are in high conformity with the doubly ionized chain end, which has also been reported for other end-functional PCLs with protonic acid [36]. Attractively, these observations not only demonstrate the negligible side reactions (e.g., transesterification and backbiting) from the 4-HBA catalyzed ROP of CL, but also further confirm the successful preparation of PCL with α-carboxyl and ω-hydroxyl chain ends.

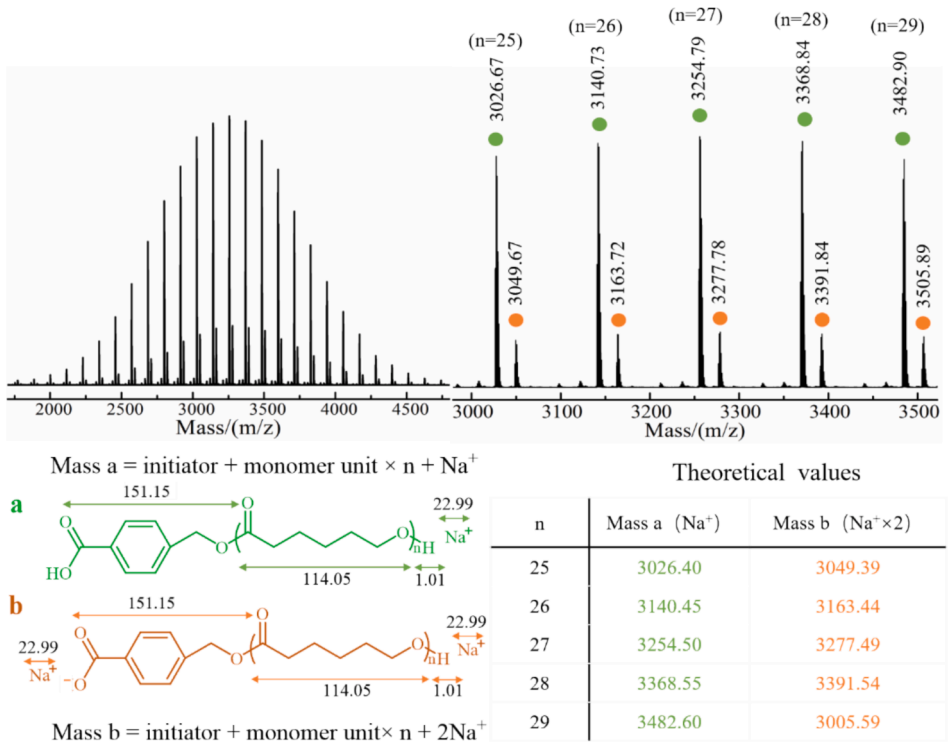
Subsequently, ^{1}H–^{13}C HMBC (Fig. 3) was carried out to further verify the chain end of the prepared PCL because it has high sensitivity to the chemical shifts of hetero-bonded nuclei, which has been widely used to illustrate the microstructure of polymers [47,48]. Apparently, there is a correlation between the signal due to the methylene protons (5.16 ppm, α-chain end) of 4-HBA and the signal attributed to the carbonyl carbon (173.50 ppm) of the PCL backbone, which undeniably identifies the connection between 4-HBA and CL unit. 2D-DOSY NMR provides a way to separate the different compounds in a mixture on the basis of the different translational diffusion coefficients (and therefore differences in the size and structure of the molecule) of each chemical species in the solution used for analysis the structure of a polymer with functional chain end [38,49]. It is noticeable that the diffusion coefficients in the ^{1}H DOSY NMR spectrum of the purified PCL (Fig. 4) recognized for signals from 4-HBA are parallel to those of the PCL backbone. This implies that 4-HBA is in connection with PCL, otherwise there should be distinguishable difference in the diffusion coefficients in the ^{1}H DOSY NMR spectrum.

It has been reported that the esterification of carboxylic acid and halogenated compound is particularly efficient in the presence of promoter 1,1,3,3-tetramethylguanidine (TMG) at room temperature [50,51]. Therefore, due to the prominent vinyl protons in the ^{1}H NMR spectrum, 4-vinylbenzyl chloride was selected to detect the benzoic acid at the a-chain end of the obtained PCL (Scheme S1). The presence of signals at 5.5 and 6.6 ppm (Fig. S5) for the purified polymer after

Table 1
ROP of CL, VL, TMC and LA initiated and catalyzed by 4-HBA.[a)]

Runs	Monomers	$[M]_0/[I]_0$[b)]	T [h]	Conv.[%][c)]	$M_{n,theo.}$ [g/mol][c)]	$M_{n,\ NMR}$ [g/mol][d)]	$M_{n,SEC}$ [g/mol][e)]	Đ[e)]
S1[f)]	CL	50	24	20.5	1 322	—-	—-	—-
S2[g)]	CL	50	24	80.7	4 758	4 400	5 300	1.17
S3	CL	50	24	96.7	5 670	4 800	6 400	1.16
S4	CL	15	8	> 99.0	1 847	1 400	2 900	1.18
S5	CL	30	8	96.5	3 456	3 800	4 800	1.14
S6	CL	100	48	95.3	11 029	12 100	13 200	1.25
S7	VL	50	16	98.5	5 083	5 500	6 800	1.13
S8	TMC	50	24	95.0	5 001	5 600	7 400	1.50
S9	LA	50	24	51.2	3 842	2 300	6 000	1.17
S10[h)]	LA	50	18	94.3	6 948	9 000	8 600	1.18
S11[i)]	LA	50	24	91.6	6 754	8 200	8 200	1.19
S12[j)]	CL	30	24	5 %	—-	—-	—-	—-
S13	CL	500	48	45.2 %	25 926	—-	28 000	1.22

a) Reaction conditions: 150 °C in bulk under CINICAT.
b) Initial molar ratio of monomer to initiator.
c) Calculated from ($[M]_0/[I]_0$) × Conv.×(MW of monomer)+(MW of CINICAT).
d) Determined by ^{1}H NMR.
e) Determined by SEC.
f) Temperature was 80°C.
g) Temperature was 130°C.
h) Temperature was 180°C.
i) Toluene was used as solvent.
j) Only benzyl alcohol as initiator.



n	Mass a (Na+)	Mass b (Na+×2)
25	3026.40	3049.39
26	3140.45	3163.44
27	3254.50	3277.49
28	3368.55	3391.54
29	3482.60	3005.59

Fig. 2. MALDI-TOF spectrum of the obtained PCL (Table 1, Table. S5). Theoretical values were calculated by the following equation: n × 114.05 + 151.15 + 1.01 + 22.99 (the main signals) and 22.99 + n × 114.05 + 151.15 + 22.99 (the minor signals), where n is the degree of polymerization and the others are the mass values corresponding to the segments/end groups comprising the PCL backbone.

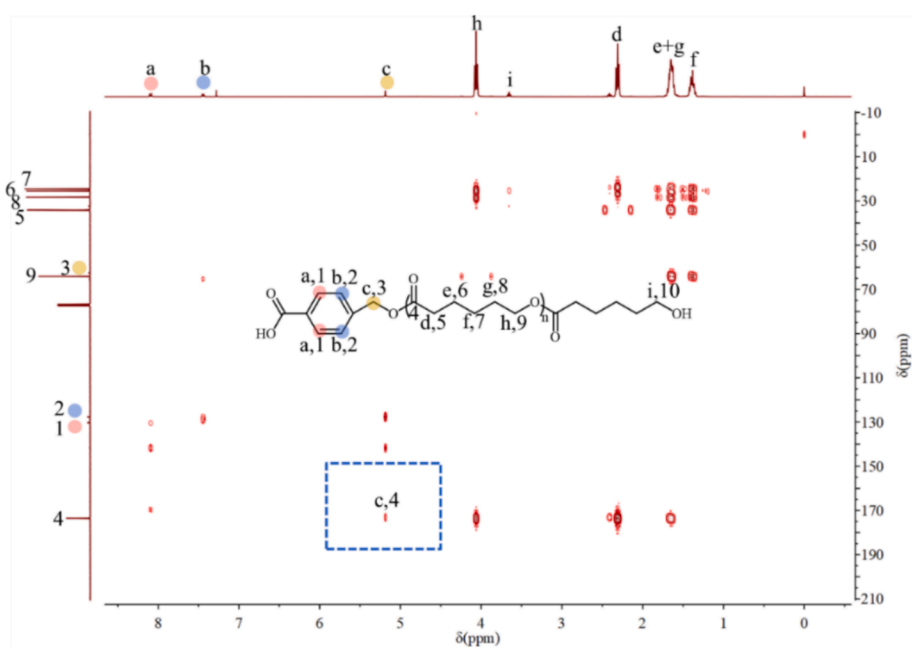


Fig. 3. ^{1}H–^{13}C HMBC spectrum ($CDCl_3$) of the isolated PCL catalyzed and initiated by 4-HBA (Table 1, Table. S5).

esterification clearly confirms the termination of benzoic acid. Interestingly, it also affords a plausible route to prepare PS-based graft copolymers by simply coupling of the benzoic acid-terminated PCL and the polymer with halogenated pendant (e.g, poly(4-vinylbenzyl chloride)). With respect to above results and discussion, we ensure that almost all PCL macromolecules were catalyzed and initiated by 4-HBA.

3.2. Polymerization details of CL in the presence of 4-HBA

It should be taken into account that 4-HBA becomes the *α*-chain end of the polymer and that free 4-HBA is almost no longer detectable after the intended polymerization time (the signals assigned to 4-HBA in ^{13}C NMR spectra before and after purification of the product are located in the same position in Fig. S6), avoiding the tedious catalyst elimination process and improving the safety of biodegradable polyesters as

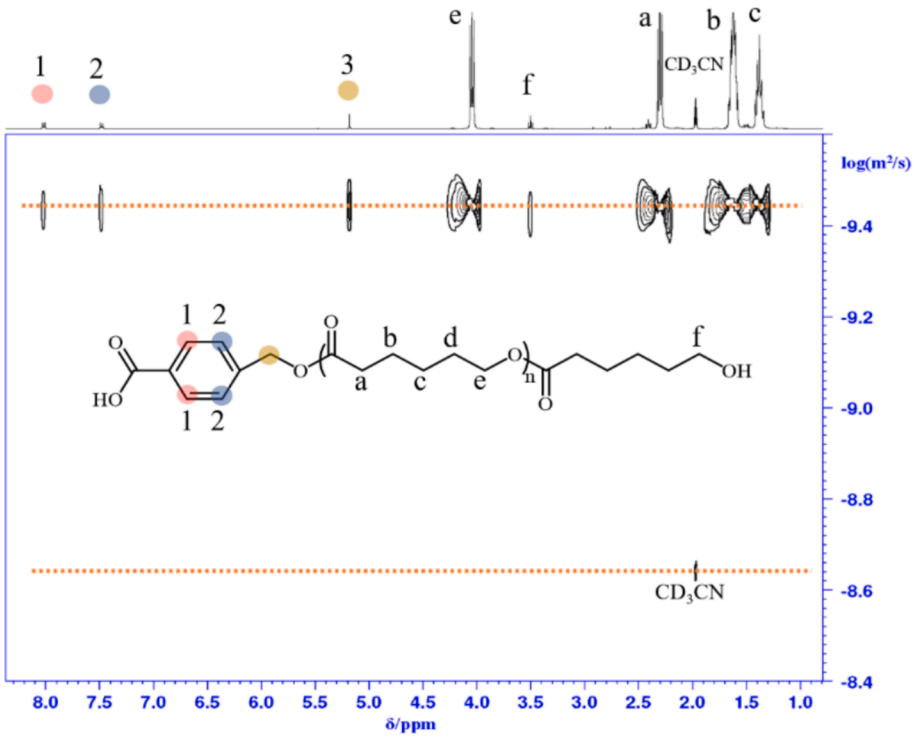


Fig. 4. ^{1}H DOSY NMR spectrum (CD_3CN) of the isolated PCL catalyzed and initiated by 4-HBA (Table 1, Table. S5).

biomaterials. We then focus on the polymerization details of CL in the presence of 4-HBA, such as the controllability, the polymerization kinetic and the chain extension, etc. It is well-known that M_n should increase with increasing initial monomer to initiator ratios to ensure a controlled/living ROP process. The M_n of the isolated PCL exhibits an increase with increasing monomer to initiator ratios (Table 1, Table. S3, S4-S6). In addition, the calculated molecular weights of $M_{n,NMR}$ are close to those of $M_{n,theo.}$ on the basis of monomer conversion, chain ends and monomer to initiator ratio (Table 1, Table. S3, S4-S6). However, the values of $M_{n,SEC}$ are slightly higher than those of $M_{n,theo.}$, which is due to the fact that the hydrodynamic volumes of the PS standards used for calibration differ significantly from those of the obtained polymers [52].

Subsequently, the kinetics of the ROP of CL catalyzed and initiated by 4-HBA were studied, with the relationship between the M_n and monomer conversion depicted in Fig. 5A. The direct proportionality observed between $M_{n,SEC}$ and monomer conversion suggests that no significant side reactions were detected for this catalytic system, a finding that is further supported by the minimal variation in $Đ$ values, which remained close to 1.1 throughout the polymerization process. In addition, the SEC profiles in Fig. 5B vividly illustrate the progressive increase in molecular weights. Fig. S7 presents a semilogarithmic plot of $\ln([CL]_0/[CL])$ against polymerization time ($[CL]_0$ and $[CL]$ represent the initial and current concentrations of the monomer), where the near-linear relationship persists up to nearly 99 % monomer conversion. Based on the above observations, a first-order kinetic and low possibility of terminal reactions could be announced associated with 4-HBA catalyzed and initiated ROP, not only indicating the constancy of the growing chains but also confirming the controlled/living character of the polymerizations [53,54].

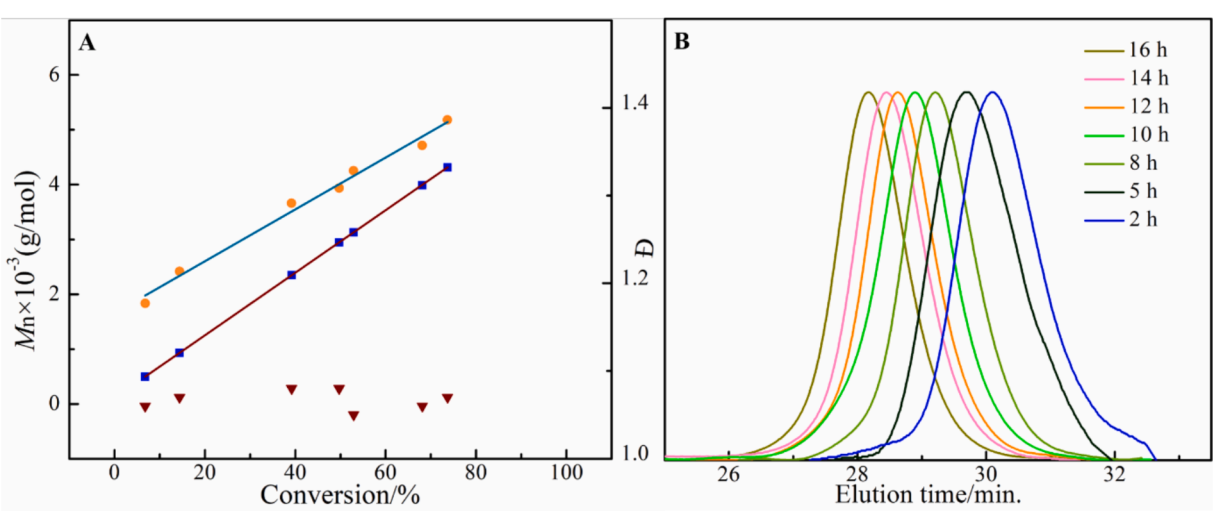


Fig. 5. (A) Dependence of $M_{n,SEC}$ (●), $M_{n,theo}$ (■) and $Đ$ (▼) of the obtained PCL on CL conversion; (B) SEC curves for the 4-HBA catalyzed and initiated ROP of CL ($[CL]_0$ = 50 mmol, $[CL]_0/[4\text{-HBA}]_0$ = 50/1).

3.3. ROP of other cyclic monomers and preparation of end-functional block copolymers in the presence of 4-HBA

In the context of the above results, it is reasonable to conclude that 4-HBA functions as a CINICAT in the ROP of CL, providing one versatile route to prepare biodegradable and biocompatible polymers with *α*-carboxyl and *ω*-hydroxyl chain ends. Consequently, a natural inquiry emerges regarding the potential application of CINICAT to other cyclic monomers, including VL, TMC and LA. Excitingly, the ROP of VL and TMC has been found to be effectively controlled, as evidenced in Table 1 (Table. S7-8). The ^{1}H and ^{13}C NMR spectra in Fig. S1(A, B) and S2(A, B) provide clear evidence of the successful incorporation of 4-HBA into PVL and PTMC. However, the monomer conversion for LA reached only 51.2 % after a 24 h polymerization period. We hypothesized that the viscosity increase of the reaction mixture due to the lower solubility of PLA (melting temperature is around 170 °C) at 150 °C and the requirement of harsh conditions due to the lower basicity of LA lead to the lower monomer conversion [38,55]. Remarkably, the monomer conversion achieved 95.6 % within 18 h by increasing the temperature to 180 °C and it also reached 91.2 % when the polymerization was conducted under reflux in toluene at 150 °C, while both polymerizations were in good control (Table 1, Table. S9-10, 1H and 13C NMR spectra in Fig. S3 (A, B)). However, when increasing the ratio of monomer to CINICAT to 500/1 (Table 1, Table S13), the obtained molar mass of PCL is 28 000 g/mol with a 45.2 % monomer conversion of CL after 48 h, indicating the limitation of CINICAT in the production of polymers with high molar mass.

Block copolymers hold substantial appeal due to their diverse applications, prompting our endeavor to synthesize end-functional, biodegradable, and biocompatible block copolymers using CINICAT. Initially, PCLs with varying molar mass (Table 2) were prepared to serve as macroinitiators. Subsequently, a secondary monomer (e.g., VL, TMC or LA) was introduced into the mixture to form the additional block of the copolymer (Scheme 2). For the purified PCL-*b*-PTMC copolymer, the peaks at 4.12 and 2.04 ppm are attributed to PTMC and the typical peaks of PCL are the same as above (Fig. S4C); for PCL-*b*-PLA, the signals at 1.55 and 5.18 ppm are assigned to the protons of PLA backbones (Fig. S4E). Moreover, the SEC traces for the copolymers depicted in Fig. S4 (B, D and F) exhibit a shift towards shorter elution times, which indicates an increase in hydrodynamic volume due to the rise in molecular weight from the macroinitiators to the resulting block copolymers [56]. Remarkably, the experimentally determined molar mass closely match the targeted values, and the dispersity are narrow, providing further evidence of the controlled/living nature of the bulk ROP catalyzed and initiated by CINICAT. However, the copolymerization of CL and LA in the presence of CINICAT was not successful, which may be due to the significantly different polymerization rates of the two monomers in this system [57,58].

Table 2
Characterization of PCL-*b*-PVL, PCL-*b*-PTMC and PCL-*b*-PLA block copolymers.[a)]

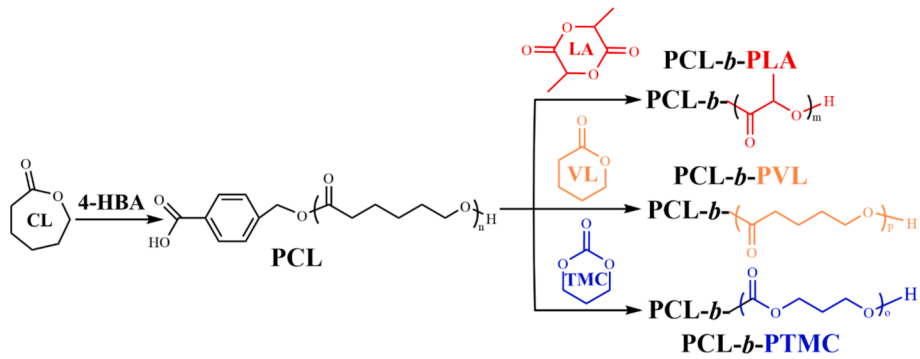
Run	Monomers	$[M]_0/[I]_0$[b)]	Conv.[%][c)]	$M_{n,SEC}$ [g/mol][d)]	$Đ$[d)]
B1[e)]	VL	50	96	16 600	1.35
B2[f)]	TMC	50	94	16 100	1.47
B3[g)]	LA	40	96	13 600	1.31

[a)] Reaction conditions: 150 °C and 13 h under bulk in the presence of 4-HBA.
[b)] Initial molar ratio of monomer to 4-HBA.
[c)] Determined by ^{1}H NMR.
[d)] Determined by SEC.
[e)] For PCL block, $M_{n,SEC}$ = 6 400 g/mol, $Đ$ = 1.18.
[f)] For PCL block, $M_{n,SEC}$ = 6 200 g/mol, $Đ$ = 1.16.
[g)] For PCL block, $M_{n,SEC}$ = 6 000 g/mol, $Đ$ = 1.17.

3.4. Evaluation of the structure of CINICAT in terms of catalytic efficiency and the possible polymerization mechanism

It is widely recognized that steric (e.g., the location of substituted group) and electronic (e.g., inductive and conjugative influences) effect significantly impact the catalytic efficiency of catalysts [59,60]. Therefore, *ortho*-(2-HBA) and *meta*-(3-HBA) hydroxymethyl substituted benzoic acid (BA) and 2-fluoro-(2-F-4-HBA) and 3-methoxy-(3-M−4−HBA) substituted HBA were chosen to compare their ROP behavior with 4-HBA (hydroxymethyl is located at the *papa*- position of BA), which could further contribute to understanding the mechanism of CINATA catalyzed and initiated ROP. The polymerization conditions and the results are summarized in Table 3. Intriguingly, the monomer conversion for the 2-HBA catalyzed polymerization was much lower than that for 3-HBA and 4-HBA (Table 3, P1-3), as shown by the ^{1}H NMR spectrum (Fig. S8) of the isolated PCL using 2-HBA displays the absence of typical signals of HBA that can be assigned to the residual water that initiated the ROP. We assume that the predominant intramolecular hydrogen bonding in 2-HBA leads to the low activation efficiency of CL by carboxylic acid since the widely accepted idea that CL is generally activated by the carboxyl group of the organocatalysts *via* an activated monomer mechanism (AMM) [61–64]. Such an assumption is well supported by the DFT calculations of the CINICAT, and Fig. 6 displays the relative free energies of the isolated reactants and the reactants complex for 2-HBA, 3-HBA and 4-HBA. Apparently, the energy for the formation of reactant complex between 2-HBA and ring-opened CL is the highest due to the formation of intramolecular hydrogen bonding (HB), while the energy for 4-HBA is the lowest, which is consistent with the experimental results (Table 3, P1-3). On the other hand, 2-F-4-HBA and 3-M−4−HBA with electron withdrawing (−F) and electron donating (−methoxy) groups were selected to study the electronic effect of CINICAT. The monomer conversion using 2-F-4-HBA is slightly lower than that of 4-HBA, while there seems an improvement with 3-M−4HBA (Table 3, P4-5), which is attributed to the improvement and reduction of acidity by substituted electron withdrawing and electron donating groups [60].

Finally, we focused on the polymerization mechanism of CL catalyzed and initiated by 4-HBA on the basis of DFT calculations, which have been widely applied to investigate the plausible mechanism of ROP with organic catalysts [21,37,65]. Considering the structural similarity of 4-HBA to the reported aromatic carboxylic acid catalyzed ROP of cyclic monomers, we envision that the AMM may be the choice [20,32,34]. Fig. 7 shows the lowest-energy pathway for the ROP of CL as well as the associated free energy of other interactions. Given the coexistence of hydroxyl groups (initiator) and carboxyl group (catalyst) groups in 4-HBA, we assume that there may be two types of initial interaction between the hydroxyl, carboxyl and CL, namely the interaction of 4-HBA and CL, and that between the two molecules of 4-HBA. As shown in Fig. 7, when the hydroxyl group of one 4-HBA is linearly connected to the carboxyl group of another 4-HBA end-to-end, with a Gibbs free energy of 1.1 kcal/mol (Path1). Nevertheless, the energy barrier of activated CL in this path is quite high (ΔG = 7.75 kcal/mol), and the further first CL ring-opening requires energy of 2.51 kcal/mol. Intriguingly, two 4-HBA molecules with hydrogen bonds between the hydroxyl group of one 4-HBA and the carboxyl group of the other arranged in parallel up and down are capable of spontaneously forming with Gibbs free energies of −5.19 kcal/mol (Path2 and 3). The energy barrier required by the carboxyl groups at both ends of the dimer 4-HBA to activate two CL molecules simultaneously is 0.96 kcal/mol (Path2), but if only one CL is activated at one end of the dimer, the reaction may be easier, due to the lower Gibbs free energy of −5.42 kcal/mol (Path3). Then, the Gibbs free energy of the first CL ring-opening in Path3 is 2.82 kcal/mol, which is still lower than that of Path2 (3.44 kcal/mol). The barrier to activation of the second CL in Path3 is 0.62 kcal/mol, but this leaves the Gibbs free energy at 14.81 kcal/mol for the open loop, suggesting that the dimer is 4-HBA in parallel up and down dimer is weak in chain growth.



Scheme 2. Synthesis of PCL-based copolymers by CINICAT.

Table 3
ROP of CL by various CINICAT.[a)]

Run	Catalysts	Conv. [%][b)]	$M_{n,NMR}$ [g/mol][b)]	$M_{n,theo.}$ [g/mol][c)]	$M_{n,SEC}$ [g/mol][d)]	$Đ$[d)]
P1	2-HBA	23.6	1 800	1 498	2 800	1.15
P2	3-HBA	95.9	5 600	5 621	7 200	1.19
P3	4-HBA	96.0	4 800	5 627	6 800	1.18
P4	2-F-4-HBA	90.3	4 500	5 320	7 100	1.26
P5	3-M−4−HBA	97.1	6 100	5 720	8 600	1.20

[a)] Reaction conditions: 150 °C in bulk, initial molar ratio of monomer to CINICAT was 50:1 and reaction time was 19 h.
[b)] Determined by ^{1}H NMR.
[c)] Calculated from $([M]_0/[I]_0) \times$ Conv.$\times$(MW of CL)+(MW of CINICAT).
[d)] Determined by SEC.

Now we examine another initial configuration, in which a 4-HBA and a CL initially interact, as shown in Path4 and Path5 in Fig. 7. Under the action of intermolecular HB, the CL molecule is activated by the carboxyl group of one 4-HBA, and the Gibbs free energy is −0.88 kcal/mol. The energy barrier of activated CL for direct ring-opening (Path4) is 6.21 kcal/mol, while activated CL interacts with another 4-HBA before ring-opening (Path5) to form the configuration in which a CL interacts with 4-HBA dimer (the same as in Path3) with a Gibbs free energy of −5.42 kcal/mol, which is obviously a more beneficial process. The subsequent reaction process of Path5 is the same as that of Path3. Activation of the second CL in Path4 appears to be easier with the Gibbs free energy of 3.08 kcal/mol. Moreover, the barrier for the ring-opening reaction of the activated second CL is also the lowest (5.58 kcal/mol). On the basis of DFT calculation, we envision that the dimeric form of 4-HBA favors the initiation of CL ring-opening due to its more stable structure and the dimeric form converts to the monomeric form during the chain growth process due to the long CL chain and high reaction temperature. Thus, the monomeric form of 4-HBA dominates to the chain growth process and the obtained PCL is incorporated with a 4-HBA.

4. Conclusions

In summary, this research successfully demonstrates the direct synthesis of α-carboxyl-ω-hydroxyl polymers through an innovative

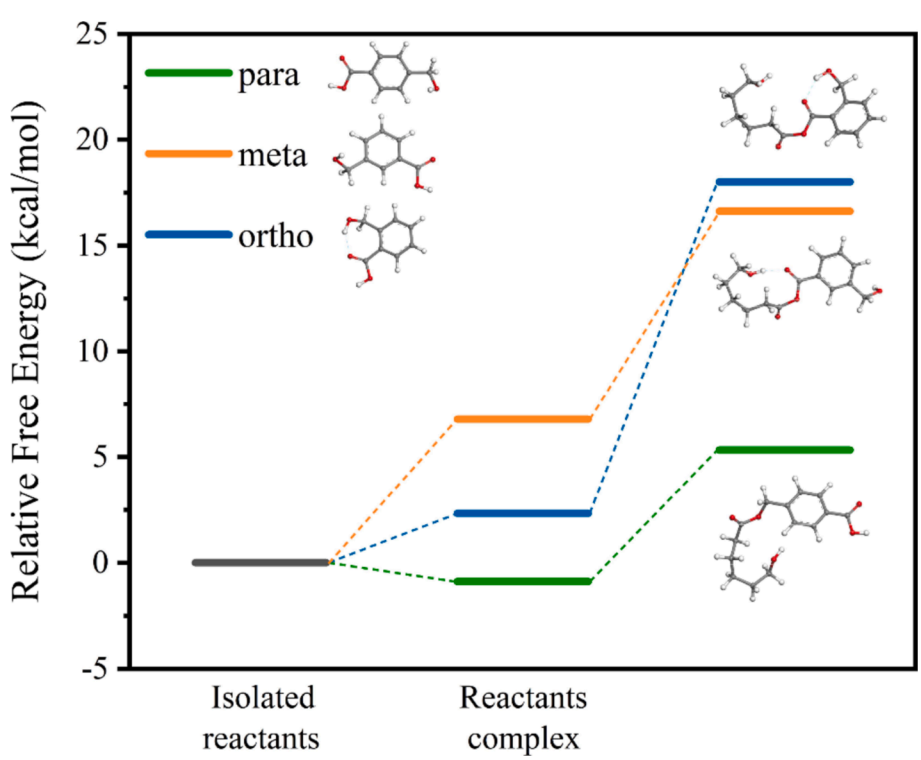


Fig. 6. DFT calculations of 2-HBA, 3-HBA and 4-HBA.

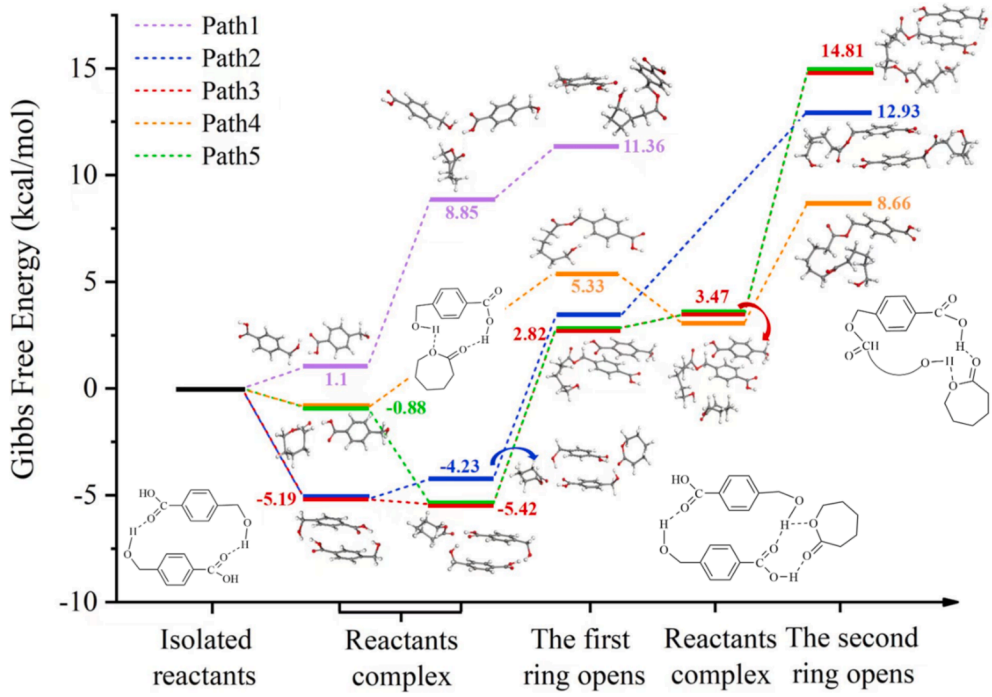


Fig. 7. Energy profile of the polymerization mechanism of CL catalyzed and initiated by 4-HBA.

approach that integrates the catalyst and initiator within a single molecule (CINICAT). 4-HBA, an organic weak acid comprising carboxyl and hydroxyl groups, was selected to proceed the ROP of various cyclic monomers, such as CL, VL, LA and TMC, as the CINICAT to give α-carboxyl-ω-hydroxyl PCL, PVL, PTMC and PLA. MALDI, 2D-NMR and other characterizations confirm the structure of the produced polymers, which have been initiated from the hydroxyl group of CINICAT. DFT calculation was carried out to study the structural influence of CINICAT on the polymerizations as well as the polymerization mechanism. It concludes that the steric effect exerts a more significant influence on the ROP polymerization of CL than the electronic effect. Furthermore, we propose a mechanism for the ROP process, where a parallel up and down dimer of 4-HBA predominantly facilitates initiation, while the monomeric form of 4-HBA mainly guides the propagation process of ROP. This study presents a significant advancement in the field of polymer chemistry, offering a straightforward and efficient method for the synthesis of α-carboxyl-ω-hydroxyl polymers. The findings have broad implications for the development of novel materials for use in medicine and other areas where biodegradability and biocompatibility are paramount.

CRediT authorship contribution statement

Jie Zhang: Methodology, Investigation, Data curation. **Siheng Zhang:** Writing – original draft, Formal analysis. **Jianda Niu:** Formal analysis. **Zhixian Dong:** Project administration. **Jinbao Xu:** Writing – original draft. **Tingting Cui:** Validation, Software. **Caihong Lei:** Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

Natural Science Foundation of Guangdong Province, China (2022A1515010624) is acknowledged.

Appendix A. Supplementary material

Supplementary data to this article can be found online at https://doi.org/10.1016/j.eurpolymj.2024.113656.

Data availability

Data will be made available on request.

References

[1] M. MacLeod, H.P.H. Arp, M.B. Tekman, A. Jahnke, The global threat from plastic pollution, Science 373 (2021) 61–65, https://doi.org/10.1126/science.abg5433.
[2] A. Stubbins, K.L. Law, S.E. Muñoz, T.S. Bianchi, L. Zhu, Plastics in the earth system, Science 373 (2021) 51–55, https://doi.org/10.1126/science.abb0354.
[3] C.M. Plummer, L. Li, Y. Chen, Ring-opening polymerization for the goal of chemically recyclable polymers, Macromolecules 56 (2023) 731–750, https://doi.org/10.1021/acs.macromol.2c01694.
[4] C. Lefay, Y. Guillaneuf, Recyclable/degradable materials *via* the insertion of labile/cleavable bonds using a comonomer approach, Prog. Polym. Sci. 147 (2023) 101764, https://doi.org/10.1016/j.progpolymsci.2023.101764.
[5] S. Saghebasl, A. Akbarzadeh, A.M. Gorabi, N. Nikzamir, M. SeyedSadjadi, E. Mostafavi, Biodegradable functional macromolecules as promising scaffolds for cardiac tissue engineering, Polym. Adv. Technol. 33 (2022) 2044–2068, https://doi.org/10.1002/pat.5669.
[6] A. Sangroniz, L. Sangroniz, S. Hamzehlou, J. del Río, A. Santamaria, J.R. Sarasua, M. Iriarte, J.R. Leiza, A. Etxeberria, lactide-caprolactone copolymers with tuneable barrier properties for packaging applications, Polymer 202 (2020) 122681, https://doi.org/10.1016/j.polymer.2020.122681.
[7] R. Xu, Y. Fang, Z. Zhang, Y. Cao, Y. Yan, L. Gan, J. Xu, G. Zhou, Recent advances in biodegradable and biocompatible synthetic polymers used in skin wound healing, Materials 16 (2023) 5459, https://doi.org/10.3390/ma16155459.
[8] B. Trzebicka, W. Wałach, P. Kurcok, M. Kowalczuk, Ring-opening (co) polymerization strategies of selected oxacyclic monomers to control and provide tailored polymers with desired properties, Polym. Adv. Technol. 35 (2024), https://doi.org/10.1002/pat.6418.
[9] C. Shi, L.T. Reilly, E.Y. Chen, Hybrid monomer design synergizing property trade-offs in developing polymers for circularity and performance, Angew. Chem. Int. Ed. Engl. 62 (2023), https://doi.org/10.1002/anie.202301850 e202301850.
[10] A. Tardy, J. Nicolas, D. Gigmes, C. Lefay, Y. Guillaneuf, Radical ring-opening polymerization: scope, limitations, and spplication to (bio)degradable materials, Chem. Rev. 117 (2017) 1319–1406, https://doi.org/10.1021/acs.chemrev.6b00319.

[11] G. Becker, F.R. Wurm, Functional biodegradable polymers via ring-opening polymerization of monomers without protective groups, Chem. Soc. Rev. 47 (2018) 7739–7782, https://doi.org/10.1039/c8cs00531a.

[12] J. Kim, H.Y. Jung, M.J. Park, End-group chemistry and junction chemistry in polymer science: Past, present, and guture, Macromolecules 53 (2020) 746–763, https://doi.org/10.1021/acs.macromol.9b02293.

[13] B. Kost, M. Basko, M. Bednarek, M. Socka, B. Kopka, G. Łapienis, T. Biela, P. Kubisa, M. Brzeziński, The influence of the functional end groups on the properties of polylactide-based materials, Prog. Polym. Sci. 130 (2022) 101556, https://doi.org/10.1016/j.progpolymsci.2022.101556.

[14] G. Pasut, F.M. Veronese, Polymer-drug conjugation, recent achievements and general strategies, Prog. Polym. Sci. 32 (2007) 933–961, https://doi.org/10.1016/j.progpolymsci.2007.05.008.

[15] J. Kopecek, Hydrophilic biomaterials: From crosslinked and self-assembled hydrogels to polymer-drug conjugates and drug-free macromolecular therapeutics, J. Control. Release 373 (2024) 1–22, https://doi.org/10.1016/j.jconrel.2024.05.012.

[16] S. Dordevic, M. Medel, J. Hillaert, E. Masia, I. Conejos-Sanchez, M.J. Vicent, Critical design strategies supporting optimized drug release from polymer-drug conjugates, Small 20 (2024), https://doi.org/10.1002/smll.202303157 e2303157.

[17] T. Saito, Y. Aizawa, T. Yamamoto, K. Tajima, T. Isono, T. Satoh, Alkali metal carboxylate as an efficient and simple catalyst for ring-opening polymerization of cyclic esters, Macromolecules 51 (2018) 689–696, https://doi.org/10.1021/acs.macromol.7b02566.

[18] M.K. Kiesewetter, E.J. Shin, J.L. Hedrick, R.M. Waymouth, Organocatalysis opportunities and challenges for polymer synthesis, Macromolecules 43 (2010) 2093–2107, https://doi.org/10.1021/ma9025948.

[19] C.-J. Zhang, X.-H. Zhang, Organocatalytic polymerization, Sci. China Chem. 62 (2019) 1087–1089, https://doi.org/10.1007/s11426-019-9483-2.

[20] J. Zhang, L. Xu, W. Xiao, Y. Chen, Z. Dong, J. Xu, C. Lei, Ring-opening polymerization of ε-caprolactone with recyclable and reusable squaric acid organocatalyst, Eur. Polym. J. 157 (2021) 110643, https://doi.org/10.1016/j.eurpolymj.2021.110643.

[21] Y. Chen, J. Zhang, W. Xiao, A. Chen, Z. Dong, J. Xu, W. Xu, C. Lei, Reinvestigation of the ring-opening polymerization of ε-caprolactone with 1,8-diazacyclo[5.4.0]undec-7-ene organocatalyst in bulk, Eur. Polym. J. 161 (2021) 110861, https://doi.org/10.1016/j.eurpolymj.2021.110861.

[22] Y. Xu, P. Guo, Z. Li, Z. Liu, T. Zhu, Y. Wang, H. Zhang, W. He, M. Lyu, K. Guo, Zwitterionic organocatalysis for ring-opening polymerization of cyclic esters, Green Chem. 26 (2024) 7825–7836, https://doi.org/10.1039/d4gc01171f.

[23] C.-G. Chae, Y.-G. Yu, H.-B. Seo, M.-J. Kim, Z. Wen, J.-S. Lee, End-capping reaction of living anionic poly(benzyl methacrylate) with a pentafluorophenyl rster for a norbornenyl-ω-rnd macromonomer with a long glexible dpacer: advantage in the well-controlled synthesis of ultrahigh-molecular-weight bottlebrush polymers, Macromolecules 52 (2019) 4828–4838, https://doi.org/10.1021/acs.macromol.9b00559.

[24] K.K. Bawa, A.M. Jazani, Z. Ye, J.K. Oh, Synthesis of degradable PLA-based diblock copolymers with dual acid/reduction-cleavable junction, Polymer 194 (2020) 122391, https://doi.org/10.1016/j.polymer.2020.122391.

[25] K. Norinkevicius, J. Madsen, L.E. Christiansen, S. Meier, A. Horner, R. Górecki, T. H.B. Regueira, A.E. Daugaard, Synthesis of poly(ethylene glycol)-*co*-poly(caprolactone) di- and triblock copolymers and effect of architecture, dispersity and end-functionalisation on their aqueous self-assembly, Polym. Chem. 14 (2023) 3433–3445, https://doi.org/10.1039/d3py00442b.

[26] P. Zhou, T. Shen, J. Ling, Synthesis and properties of polypeptoid-containing block copolymers: a review, J. Polym. Sci. 59 (2021) 2946–2958, https://doi.org/10.1002/pol.20210507.

[27] P. Zhang, V. Ladelta, N. Hadjichristidis, Living/controlled anionic polymerization of glycolide in fluoroalcohols: toward sustainable bioplastics, J. Am. Chem. Soc. 145 (2023) 14756–14765, https://doi.org/10.1021/jacs.3c03253.

[28] J. Xu, Y. Chen, W. Xiao, J. Zhang, M. Bu, X. Zhang, C. Lei, Studying the ring-opening polymerization of 1,5-fioxepan-2-one with organocatalysts, Polymers 11 (2019) 1642–1653, https://doi.org/10.3390/polym11101642.

[29] W.D. Fu, J. Jiang, Y. Zhang, J.-J. Li, L. Zhao, Z. Xi, Kinetic modeling of cationic ring-opening polymerization for the synthesis of biodegradable poly(ε-caprolactone), Chem. Eng. Sci. 290 (2024) 119876, https://doi.org/10.1016/j.ces.2024.119876.

[30] Y. Chen, P. Zhang, S. Liu, D. Pahovnik, E. Žagar, J. Zhao, G. Zhang, Noncovalent protection for direct synthesis of α-amino-ω-hydroxyl poly(ethylene oxide), ACS Macro Lett. 10 (2021) 737–743, https://doi.org/10.1021/acsmacrolett.1c00316.

[31] E. Wojtczak, P. Kubisa, M. Bednarek, Thermal stability of polylactide with different end-groups depending on the catalyst used for the polymerization, Polym. Degrad. Stab. 151 (2018) 100–104, https://doi.org/10.1016/j.polymdegradstab.2018.03.003.

[32] L. Mezzasalma, J. De Winter, D. Taton, O. Coulembier, Benzoic acid-organocatalyzed ring-opening (co)polymerization (ORO(c)P) of l-lactide and ε-caprolactone under solvent-free conditions: from simplicity to recyclability, Green Chem. 20 (2018) 5385–5396, https://doi.org/10.1039/c8gc03096k.

[33] L. Mezzasalma, S. Harrisson, S. Saba, P. Loyer, O. Coulembier, D. Taton, Bulk organocatalytic synthetic access to statistical copolyesters from l-Lactide and epsilon-caprolactone using benzoic acid, Biomacromolecules 20 (2019) 1965–1974, https://doi.org/10.1021/acs.biomac.9b00190.

[34] J. Xu, J. Song, S. Pispas, G. Zhang, Controlled/living ring-opening polymerization of ε-caprolactone with salicylic acid as the organocatalyst, J. Polym. Sci. A Polym. Chem. 52 (2014) 1185–1192, https://doi.org/10.1002/pola.27104.

[35] J. Liu, L. Liu, Ring-opening polymerization of ε-caprolactone initiated by natural amino acids, Macromolecules 37 (2004) 2674–2676, https://doi.org/10.1021/ma0348066.

[36] K. Kaluzynski, J. Pretula, P. Lewinski, S. Kaźmierski, S. Penczek, Catalysis in polymerization of cyclic esters. Catalyst and initiator in one molecule. Polymerization of ε-caprolactone, J. Catal. 392 (2020) 97–107, https://doi.org/10.1016/j.jcat.2020.09.026.

[37] K. Kaluzynski, J. Pretula, S. Kaźmierski, P. Lewinski, M. Cypryk, S. Penczek, Cationic polymerization of cyclic trimethylene carbonate induced with initiator and catalyst in one molecule: polymer structure, kinetics and DFT, J. Catal. 415 (2022) 200–217, https://doi.org/10.1016/j.jcat.2022.10.009.

[38] P. Lewinski, K. Kaluzynski, J. Pretula, G. Mielniczak, S. Penczek, Catalysis in polymerization of cyclic esters. Catalyst and initiator in one molecule. Polymerization of lactide, J. Catal. 405 (2022) 249–264, https://doi.org/10.1016/j.jcat.2021.11.038.

[39] K. Kaluzynski, J. Pretula, P. Lewinski, S. Kaźmierski, S. Penczek, Synthesis and properties of functionalized poly(ε-caprolactone); chain polymerization followed by polycondensation in one pot with initiator and vatalyst in one molecule, Synthesis and Molecular Structures, Macromolecules 55 (2022) 2210–2221, https://doi.org/10.1021/acs.macromol.1c02325.

[40] Y. Li, M. Wu, D. Chen, L. Jiang, J. He, Z. Luo, W. Wang, J. Wang, One-step highly selective oxidation of p-xylene to 4-hydroxymethylbenzoic acid over Cu-MOF catalysts under mild conditions, Mol. Cat. 477 (2019) 110542, https://doi.org/10.1016/j.mcat.2019.110542.

[41] B. Delley, An all-electron numerical method for solving the local density functional for polyatomic molecules, J. Chem. Phys. 92 (1990) 508–517, https://doi.org/10.1063/1.458452.

[42] J.P. Perdew, K. Burke, M. Ernzerhof, Generalized gradient approximation made simple, PhysRevLett. 77 (1996) 3865–3868, https://doi.org/10.1103/PhysRevLett.77.3865.

[43] J. Xu, J. Song, S. Pispas, G. Zhang, Metal-free controlled ring-opening polymerization of ε-caprolactone in bulk using tris(pentafluorophenyl)borane as a catalyst, Polym. Chem. 5 (2014) 4726–4733, https://doi.org/10.1039/c4py00342j.

[44] L. Mezzasalma, A.P. Dove, O. Coulembier, Organocatalytic ring-opening polymerization of l-lactide in bulk: a long standing challenge, Eur. Polym. J. 95 (2017) 628–634, https://doi.org/10.1016/j.eurpolymj.2017.05.013.

[45] J. Drzeżdżon, D. Jacewicz, A. Sielicka, L. Chmurzyński, MALDI-MS for polymer characterization-Recent developments and future prospects, TrAC, Trends Anal. Chem. 115 (2019) 121–128, https://doi.org/10.1016/j.trac.2019.04.004.

[46] Y. Park, S. Jin, I. Noda, Y.M. Jung, Continuing progress in the field of two-dimensional correlation spectroscopy (2D-COS): part III Versatile Applications, Spectrochim. Acta A 284 (2023) 121636, https://doi.org/10.1016/j.saa.2022.121636.

[47] X. Xing, K. Xing, Y.S.Y. Hsieh, D.W. Abbott, Inequality relations for NMR-based polymer homoblock analysis and extended application: reanalysis of historical data on alginates, chitosans, homogalacturonans, and galactomannans, Carbohydr. Res. (2024) 109189, https://doi.org/10.1016/j.carres.2024.109189.

[48] X. Zhang, Y. Xia, Y. Sun, C. Zhang, X. Zhang, Water-degradable oxygen-rich polymers with AB/ABB units from fast and selective copolymerization, Angew. Chem. Int. Ed. Engl. 63 (2024), https://doi.org/10.1002/anie.202315524 e202315524.

[49] P. Lewinski, S. Sosnowski, S. Kazmierski, S. Penczek, l-Lactide polymerization studied by 1H NMR with diffusion-ordered spectroscopy (DOSY): a "One NMR Tube Experiment" providing data on monomer conversion, polymer structure, M_n and M_w, Polym. Chem. 6 (2015) 4353–4357, https://doi.org/10.1039/c5py00455a.

[50] Q. Li, T. Wang, C. Ma, W. Bai, R. Bai, Facile and highly efficient strategy for synthesis of functional polyesters via tetramethyl guanidine promoted polyesterification at room temperature, ACS Macro Lett. 3 (2014) 1161–1164, https://doi.org/10.1021/mz5005184.

[51] J. Xu, W. Liang, J. Zhang, Z. Dong, C. Lei, Synthesis of side-chain functional poly(ε-caprolactone) via the versatile and robust organo-promoted esterification eeaction, Eur. Polym. J. 179 (2022) 111526, https://doi.org/10.1016/j.eurpolymj.2022.111526.

[52] B. Kost, M. Socka, K. Cichoń, M. Brzeziński, M. Basko, T. Biela, Molar mass determination *via* linear regression from SEC measurements: the case of polylactide, polycaprolactone and poly(2,2-dimethylthrimethylene carbonate), Polym. Adv. Technol. 35 (2024) 1, https://doi.org/10.1002/pat.6296.

[53] S. Xiang, J. Zhang, S. Liu, P. Braunstein, Z. Li, Aluminum complexes bearing aminoquinoline ligands with different pendant donors: Remarkable impact of hemilability on the ring-opening polymerization of ε-Caprolactone, Macromolecules 57 (2024) 3604–3613, https://doi.org/10.1021/acs.macromol.4c00120.

[54] M. Castillo-Santillan, J.R. Torres-Lubian, A. Martínez-Richa, S.T. Huerta-Marcial, M.C. Gutierrez, K. Loos, M.G. Peréz-García, J.D. Mota-Morales, From polymer blends to a block copolymer: ring-opening polymerization of L-lactide/ε-caprolactone eutectic system, Polymer 262 (2022) 125432, https://doi.org/10.1016/j.polymer.2022.125432.

[55] S. Asano, J. Choi, T.H. Tran, N. Chanthaset, H. Ajiro, The influence of chain-end functionalization and stereocomplexation on the degradation stability under alkaline condition, Polym. Adv. Technol. 33 (2021) 991–999, https://doi.org/10.1002/pat.5573.

[56] Z. Xu, Q. Dong, W. Li, Architectural design of block copolymers, Macromolecules 57 (2024) 1869–1884, https://doi.org/10.1021/acs.macromol.3c01730.

[57] D. Dakshinamoorthy, F. Peruch, Block and random copolymerization of ε-caprolactone, L-, and rac-lactide using titanium complex derived from aminodiol ligand, J. Polym. Sci. A Polym. Chem. 50 (2012) 2161–2171, https://doi.org/10.1002/pola.25983.
[58] M. Florczak, J. Libiszowski, J. Mosnacek, A. Duda, S. Penczek, L,L-Lactide and ε-caprolactone block copolymers by a 'poly(L,L-lactide) block first' route, Macromol. Rapid Comm. 28 (2007) 1385–1391, https://doi.org/10.1002/marc.200700160.
[59] J.P. Wagner, P.R. Schreiner, London dispersion in molecular chemistry–reconsidering steric effects, Angew. Chem. Int. Ed. Engl. 54 (2015) 12274–12296, https://doi.org/10.1002/anie.201503476.
[60] L. Cavallo, A. Correa, C. Costabile, H. Jacobsen, Steric and electronic effects in the bonding of N-heterocyclic ligands to transition metals, J. Organomet. Chem. 690 (2005) 5407–5413, https://doi.org/10.1016/j.jorganchem.2005.07.012.
[61] S. Penczek, Cationic ring-opening polymerization (CROP) major mechanistic phenomena, J. Polym. Sci. A Polym. Chem. 38 (2000) 1919–1933, https://doi.org/10.1002/(SICI)1099-0518(20000601)38:11.
[62] N.E. Kamber, W. Jeong, R.M. Waymouth, R.C. Pratt, B.G.G. Lohmeijer, J. L. Hedrick, Organocatalytic ring-opening polymerization, Chem. Rev. 107 (2007) 5813–5840, https://doi.org/10.1021/cr068415b.
[63] P. Kubisa, S. Penczek, Cationic activated monomer polymerization of heterocyclic monomers, Prog. Polym. Sci. 24 (1999) 1409–1437, https://doi.org/10.1016/S0079-6700(99)00028-3.
[64] P. Lewinski, J. Pretula, K. Kaluzynski, S. Kaźmierski, S. Penczek, ε-Caprolactone: activated monomer polymerization; controversy over the mechanism of polymerization catalyzed by phosphorus acids (diarylhydrogen phosphates). Do acids also act as initiators? J. Catal. 371 (2019) 305–312, https://doi.org/10.1016/j.jcat.2019.02.015.
[65] I. Nifant'ev, P. Ivchenko, DFT modeling of organocatalytic ring-opening polymerization of cyclic esters: a crucial role of proton exchange and hydrogen bonding, Polymers 11 (2019) 2078, https://doi.org/10.3390/polym11122078.